Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/28/2006
PI Name:	Kamotani, Yasuhiro Ph.D.		
Project Title:	Gas Evolution Effect on Mass Transfer in Rotating Electrochemical Cells Under Microgravity Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	yxk@case.edu	**Fax:**	FY 216 368-6445
PI Organization Type:	UNIVERSITY	**Phone:**	216 368-6455
Organization Name:	Case Western Reserve University		
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City:	Cleveland	**State:**	OH
Zip Code:	44106	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	02/21/2001	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Green, Robert D	**Contact Phone:**	216-433-5402
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Flight Program:			
Flight Assignment:	Note: date extended to 11/30/2004		
Key Personnel Changes/Previous PI:	Kamotani, Yasuhiro		
COI Name (Institution):			
Grant/Contract No.:	NCC3-823		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Task Abstract/Description: The Principal Investigator will perform: 1) A ground-based experimental study of bubble evolution effect on the mass transfer in a rotating electrochemical cell, and 2) parabolic flight tests on bubble evolution in an electrochemical cell under reduced gravity condition. Task Objective:The objective of this work is to investigate rotating electrochemical cells as a means of enhancing mass transfer significantly and, at the same time, removing the generated bubbles effectively, in microgravity as well as in normal gravity. Task Significance:On earth, gas evolution in electrochemical reactors occurs widely in the electrochemical industry. In micrigravity, gas bubbles generated in an electrochemical cell must be removed from the electrode surfaces by forced		

	convection of the electrolyte. Moreover, in many applications an additional separator is needed to remove the bubbles from the electrolyte. By rotating the cells one can remove the bubbles from the electrodes and the electrolyte.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 03/06/2024)